Condition-Blind by Design

Michigan's Act 51 Formula: Does Road Funding Reach the Roads That Need It Most?

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Abstract

Michigan distributes about $1.3 billion a year to county road agencies through a statutory formula that weights vehicle registrations, road mileage, and population, and that contains no measure of road condition. I examine whether the resulting allocations nevertheless correspond to road quality. Combining an original county-month funding panel constructed from 43 Michigan Department of Transportation payment PDFs with segment-level pavement roughness from the state's biennial condition survey, I find no statistically significant relationship between funding per road-mile and road roughness ($r = -0.020$, $p = 0.92$). By contrast, the formula's inputs (registrations, population, and mileage) explain 93% of the variation in funding per road-mile. Funding per road-mile varies 5.4-fold across counties, so the allocation is far from uniform; it simply varies along a dimension other than condition. These findings are descriptive and do not establish whether the formula causes disparities in road conditions or whether alternative allocation methods would improve road quality. This study concludes by outlining what the cross-section cannot identify.

1. Introduction

Road maintenance is the part of infrastructure policy where the gap between spending and outcomes is easiest to see and hardest to explain. The American Society of Civil Engineers (2021) graded the nation's roads a D, and deferred maintenance imposes measurable costs on the households and firms that use the network. Bock, Cardazzi and Humphreys (2025) find that pavement damage reduces both traffic speed and safety, and Gertler et al. (2024) show that maintenance spending on Indonesian highways produced local economic gains. If maintenance money has returns, then where it goes should matter.

A growing literature asks what actually determines that allocation. Alm and Dronyk-Trosper (2021) examine the drivers of road infrastructure spending; Kim (2025) compares funding mechanisms for state road maintenance; Kim and Ebdon (2021) document how thinly U.S. counties practice formal asset management. A parallel public-finance literature going back to Buchanan (1963) asks what earmarking does to budget behavior, with Nguyen-Hoang and Duncombe (2012) showing that earmarked highway revenues transmit their own volatility to spending. What this work largely takes as given is the allocation rule itself.

Michigan offers an unusually clean case for looking at the rule directly. Public Act 51 of 1951 fixes in statute how the Michigan Transportation Fund is divided among the state trunkline system, county road agencies, and cities and villages, and then fixes how the county share is split across all 83 counties. The county formula weights resident vehicle registrations, road mileage of four kinds, population outside incorporated municipalities, an equal per-county share, and a snow factor. It contains no measure of pavement condition (Citizens Research Council of Michigan, 2022, 2024).

This paper asks a narrow empirical question that follows from that design: does the money nonetheless reach the roads that need it most? A formula can be condition-blind and still track need if its inputs happen to correlate with deterioration. For instance, rural counties with heavy truck traffic and long mileage might draw more per mile. The question is whether they do. The funding panel does not exist in machine-readable

form; Michigan publishes it as monthly PDF payment sheets. I wrote a parser that recovers 29,376 county line items from 43 such sheets and reconciles 99.8 percent of them against the totals printed on the documents themselves.

Across the 29 counties for which a matched 2023-to-2025 pavement panel exists, funding per road-mile shows no statistically significant relationship with baseline road roughness ($r = -0.020$, $p = 0.92$). It is instead closely tied to the formula's own inputs: registrations, population, and mileage together account for 93% of the variation in funding per road-mile. The allocation is far from arbitrary, since it varies 5.4-fold across counties, but it varies along a dimension other than condition: registrations and urbanization.

2. A formula that cannot see condition

The Michigan Transportation Fund collects fuel taxes and vehicle registration fees and distributes them under Act 51. The top-level split sends 39.1 percent to the state trunkline fund, 39.1 percent to county road agencies and 21.8 percent to cities and villages. The county share is then divided among the 83 counties by a fixed set of weights, reproduced in Table 1.

Table 1. Statutory allocation factors for the county share of the Michigan Transportation Fund

Allocation factor	Weight
Resident vehicle registrations	47.90%
County local road mileage	16.41%
County road mileage in urban areas	9.90%
Equal distribution across the 83 counties	9.58%
Population outside incorporated municipalities	8.84%
County primary road mileage	6.39%
Snow formula	1.00%
Pavement condition	*none*

Each county also receives a $10,000 baseline before the formula is applied. Source: Citizens Research Council of Michigan (2024).

Two features of Table 1 are noteworthy. First, nearly half the county share moves with registrations, a variable that tracks population and vehicle ownership rather than road wear. Second, road condition does not appear. As such, the formula cannot respond to a county whose pavement is failing, and it cannot stop paying a county whose pavement is sound. These allocation factors were established as proxies for funding needs in 1951, rather than being based on direct measures of road condition. Citizens Research Council of Michigan, 2024)

3. Data

Funding. MDOT publishes county distributions as monthly "County Continuation Sheet" PDFs: one-page blocks per county, each listing eight to eleven line items and a printed total. There is no machine-readable release. I wrote a parser that reads these sheets by page geometry rather than by pattern-matching raw text, because each county's mailing address shares a text line with its line-item labels and splitting those lines on whitespace is ambiguous. Across 43 sheets covering June 2022 through December 2025, it recovers 29,376

county line items for all 83 counties, totaling $1.27 billion distributed in 2023, $1.30 billion in 2024, and $1.29 billion in 2025.

Each county's extracted line items are summed and compared against the total printed on the sheet, and 3,562 of 3,569 county-months reconcile exactly. Three of the 43 months are scanned images with no text layer; those are rendered and read with optical character recognition through the same geometry logic, and the seven remaining failures are all in those scanned months or in one anomalous September 2025 sheet whose county blocks appear out of order. Every failure is reported by county, month, and dollar gap rather than silently absorbed. As an external check, the output was compared cell by cell against an earlier hand-checked build of the same panel: of 41,832 overlapping line-item cells, 41,816 agree to the cent, and every disagreement falls in the scanned months.

Pavement condition. Road roughness comes from MDOT's Pavement Condition Measures layer, which reports the International Roughness Index for individual road segments. Michigan surveys on a biennial rotation by functional class: odd years cover principal arterials, while even years cover minor arterials and major collectors. Consequently, comparing a county's mean IRI across adjacent years measures a change in which roads were surveyed, not a change in the roads. Only 2023 and 2025 form a matched before-and-after panel of the same segments, which is the window used here.

Restricting to county-owned roads and matching segments across the two surveys yields 7,042 segments in 29 counties. The federal-aid reporting basis means true local roads are effectively absent: 6,039 segments are principal arterials, 677 minor arterials, 301 major collectors, and 25 minor collectors. Coverage is therefore a real limitation, as discussed in Section 6.

County controls. Road mileage by class, population outside incorporated municipalities, and trailing twelve-month vehicle-registration revenue come from MDOT's Act 51 population-and-mileage report. Total county road mileage is the denominator used to normalize funding; registration revenue measures the formula's largest input, not a second funding term, since it is mechanically a component of the allocation.

4. Empirical approach

Act 51 sets funding at the county level, so every segment within a county carries the same funding value. The effective sample for any question about funding is therefore 29 counties, not 7,042 segments. I run the analysis on county-level collapses of the segment data. This same fact rules out county fixed effects: funding is constant within a county, so a full set of county indicators would absorb the funding variable exactly and leave nothing to estimate. It also disciplines inference, as I discuss at the end of the section. With 29 observations, I use heteroskedasticity-robust standard errors and interpret the results through confidence intervals rather than relying on large-sample approximations.

The question this section answers is whether the allocation formula directs money toward counties whose roads are in worse condition. Let i index counties, $i = 1, \ldots, 29$. Let f_i denote the Act 51 funding county i received per road-mile over the lagged window, expressed in thousands of dollars, and let $need_i$ denote a measure of the county's baseline road condition, defined so that larger values mean worse roads: either mean baseline IRI (roughness in inches per mile) or the share of the county's rated segments in Poor condition at baseline. The baseline specification is

$$f_i = \beta_0 + \beta_1 \, need_i + \varepsilon_i, \quad (1)$$

estimated by ordinary least squares with Huber–White (HC1) heteroskedasticity-robust standard errors. The coefficient of interest is β_1, the conditional-mean change in funding per road-mile associated with a one-unit increase in need. The sign of β_1 is what makes equation (1) a test of the statute's design. If the formula were responsive to condition, if worse-off counties received more money per mile, then $\beta_1 > 0$. The statute instead allocates based on vehicle registrations, population, and mileage, with no condition term, which implies $\beta_1 = 0$. The null H_0: $\beta_1 = 0$ is thus the hypothesis that funding is unrelated to road need, and it is the hypothesis the statute itself predicts.

Because the formula's inputs are known, I can go past the bivariate relationship and ask whether any responsiveness to condition survives once the formula's genuine drivers are held fixed. Registrations and urban mileage both correlate with condition, so a raw β_1 near zero could in principle mask offsetting relationships. I therefore estimate the augmented specification

$$f_i = \beta_0 + \beta_1 \, need_i + \beta_2 \, urban_i + \beta_3 \, UP_i + \varepsilon_i, \quad (2)$$

where $urban_i$ is the share of the county's road mileage classified as urban, and UP_i indicates an Upper Peninsula County. These are the observable axes along which the formula distributes money, so they are the appropriate conditioning set. A β_1 that remains near zero in (2) is a stronger statement than in (1): it says the formula ignores the condition even among counties that are otherwise alike in urbanization and geography.

5. Results

Table 2 gives the county-level descriptives. The dispersion in funding is the first thing worth noting: the best-funded county receives 5.4 times as much per road-mile as the least-funded. Whatever the formula is doing, it is not distributing evenly.

Table 2. County-level descriptive statistics (29 counties, 2023–2025 window)

	Mean	SD	Min	Max
Act 51 funding per road-mile	$26,499	$13,422	$14,083	$76,233
Mean baseline IRI (in./mile)	150.4	—	53	247
Share of segments rated Poor	16.4%	—	0%	100%
Matched segments per county	243	—	2	2,004

Funding is total Act 51 receipts over the lagged window divided by total county road miles. IRI is the International Roughness Index; higher values are rougher roads. Segment counts are highly skewed, which Section 6 returns to.

The question is whether Act 51 funding reaches the counties whose roads are in the worst condition. Table 3 reports the correlations that answer it: funding per road-mile shows no statistically significant correlation with either measure of need, while it is strongly and significantly correlated with each of the formula's inputs.

Table 3. Correlation of funding per road-mile with need and with formula inputs

	Pearson r	p	Spearman ρ	p
Measures of need				
Mean baseline IRI	−0.020	0.918	−0.115	0.553

	Pearson r	p	Spearman ρ	p
Measures of need				
Lane-mile-weighted baseline IRI	−0.018	0.928	—	—
Share of segments rated Poor	−0.146	0.449	+0.139	0.472
Formula inputs				
Vehicle-registration revenue	+0.945	<0.001	+0.926	<0.001
Population outside incorporated areas	+0.844	<0.001	+0.899	<0.001
Urban road share	+0.811	<0.001	+0.907	<0.001
Total county road miles	+0.757	<0.001	+0.838	<0.001

N = 29 counties. Note that the Pearson and Spearman coefficients for the Poor share carry opposite signs, which is itself evidence that there is no stable relationship to detect.

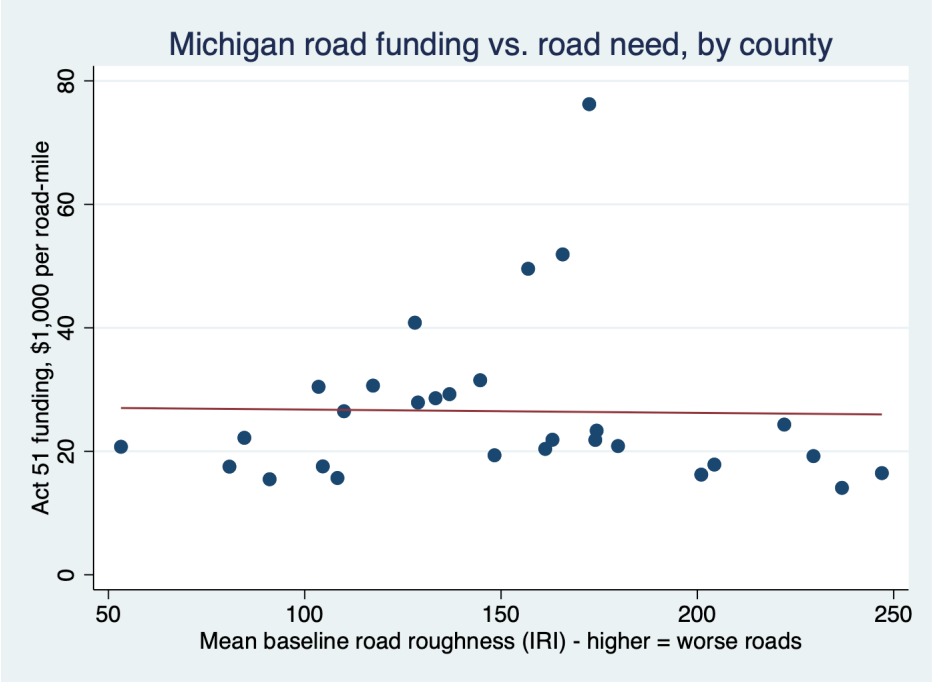


Figure 1. Act 51 funding per road-mile against mean baseline road roughness, by county. The fitted line is flat: slope −0.005 (robust SE 0.037), R-squared 0.0004.

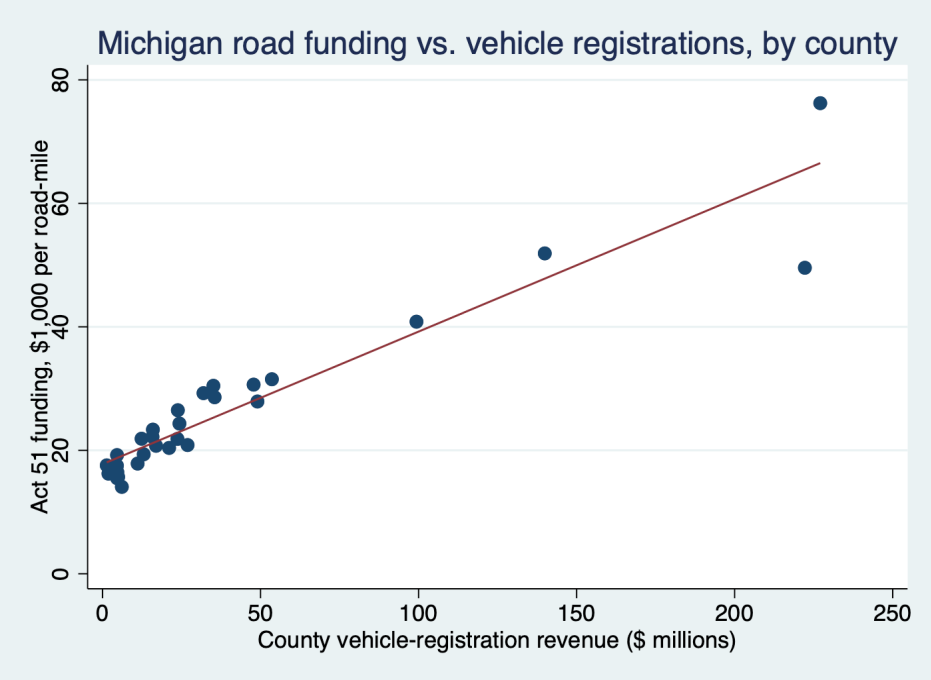


Figure 2. The same funding measure against county vehicle-registration revenue, the formula's largest single input. Slope +0.215 (robust SE 0.037), R-squared 0.894.

Table 4 presents the regression results. Column 1 regresses funding per road-mile on mean baseline IRI and reproduces the flat line of Figure 1: the coefficient is −0.005 (p = 0.89), and the regression explains essentially none of the variation in funding (R^2 = 0.0004). The magnitude is as telling as the significance. Taken at face value, the point estimate implies that moving across the entire observed range of county road roughness, from the smoothest county at 53 IRI to the roughest at 247, changes predicted funding by about $1,000 per mile, against a mean near $50,000 and a spread of more than fivefold. What separates a well-paved county from a failing one is not how much Act 51 money arrives. Column 2 substitutes the share of a county's segments rated Poor and returns a coefficient that is negative and marginally significant (−0.081, p = 0.07): each additional percentage point of Poor pavement is associated with roughly $80 less funding per mile. Although the sample is small and the estimate is descriptive, it points away from need-based targeting. Column 3 adds the two confounders the formula's structure makes obvious, urban road share and an Upper Peninsula indicator, and the condition coefficient stays indistinguishable from zero (0.017, p = 0.55) while urban road share does all the explanatory work essentially, lifting the R^2 from near zero to 0.68. Columns 4 and 5 show the positive-space counterpart: registration revenue alone explains 89 percent of the variation in funding per road-mile, and the three main formula inputs together explain 93 percent.

Two features of these results deserve emphasis: what they show is a genuine and quantitatively large mismatch: funding is highly predictable, nine-tenths of its cross-county variance is pinned down by registrations alone, but predictable from inputs that carry no information about pavement condition. The flat condition relationship is therefore not the artifact of a variable that fails to vary; Columns 4 and 5 confirm that funding varies enormously. It simply lands on a different axis. What the results do not show is equally important. With 29 counties, the null on condition is loosely bounded: the 95 percent interval on the Column 1 slope admits, across the full condition range, anything from about $15,600 less to $13,600 more per mile for the worst-condition counties, so the data are consistent with no relationship and cannot exclude a modest one in either direction. I therefore do not claim that funding is unrelated to need, only that no such relationship appears in these data; and that if it exists, it would be small. These columns only describe how money is allocated, and the two competing readings of the funding–condition association are separated only by the change model of Section 7. Finally, the 29 counties are those with federal-aid

pavement coverage, so the picture is of the state's more heavily traveled county roads rather than its network.

Table 4. OLS regressions of funding per road-mile ($000s) on condition and formula inputs

	(1)	(2)	(3)	(4)	(5)
Mean baseline IRI	−0.005		+0.017		
	(0.037)		(0.027)		
Share rated Poor		−0.081			
		(0.043)			
Urban road share			+81.63***		
			(18.17)		
Upper Peninsula			+5.78		
			(3.72)		
Registration revenue ($M)				+0.215***	+0.215***
				(0.037)	(0.050)
Population, road miles	no	no	no	no	yes
R-squared	0.0004	0.021	0.678	0.894	0.928
Counties	29	29	29	29	29

Heteroskedasticity-robust (HC1) standard errors in parentheses. *** p < 0.01. The dependent variable is Act 51 funding per county road-mile in thousands of dollars.

6. What this design cannot establish

Like all studies, this study has limitations. Three limitations are discussed below.

The tercile comparison is fragile. Sorting counties into terciles of baseline condition and comparing mean funding is the most intuitive way to present this finding, and it is the one I would drop: With 29 counties, assigning a single county to a bin changes the answer. Under one standard binning rule, the worst-condition tercile receives $25,048 per road-mile against $23,760 for the best, a difference of 5.4 percent (Welch t = 0.20, p = 0.84). Under the binning rule Stata's xtile applies, Wayne County shifts from the worst tercile to the middle, and the same comparison reverses to $19,360 against $23,760. Both are defensible implementations; neither is a finding because a result that reverses on the placement of a single county reflects the binning rule rather than any fact about the data. The correlation and regression results in Tables 3 and 4 do not depend on binning, which is why they carry the argument.

Segment counts are uneven. Matched segments per county range from 2 to 2,004, with a median of 51. County mean IRI is therefore measured much more precisely for some counties than others, and the county share rated Poor takes the value 0 or 100 percent in the smallest counties. Restricting to the fifteen counties with at least fifty matched segments leaves the null intact but with unstable signs (IRI r = +0.21, p = 0.46; Poor share r = −0.21, p = 0.46). Weighting county IRI by lane-miles rather than taking a simple mean does not meaningfully change the results (r = −0.018). Running the same regression on all 7,042 segments with standard errors clustered by county produces a small positive and nominally significant coefficient, but that

specification effectively weights counties by how many segments they contain, which is a proxy for county size and therefore for the formula inputs themselves, which answers a different question.

The cross-section cannot identify effects. Counties receiving more per mile do have better roads in these data: sorting instead by funding, the highest-funded third averages 138.8 IRI and 9.0 percent of segments Poor, against 165.2 and 25.6 percent for the lowest-funded third. That pattern has at least two readings. Money may have improved those roads, or those counties may be urban, dense, and differently maintained for reasons that have nothing to do with Act 51. A single cross-section cannot separate the two, and it would be a mistake to read the association as a return to spending. Separating them is the purpose of the change model described next. Coverage is a further caveat: the pavement layer reports federal-aid roads, so the 29 counties here are not a random sample of Michigan's 83, and true local roads are absent.

7. Ongoing work

The natural next question is whether the funding that does arrive improves pavement, and that requires a design this cross-section cannot supply. I am extending the data to a segment-level first-difference model in which the change in IRI over a survey window is regressed on lagged county funding per road-mile, controlling for baseline roughness, truck and total traffic, functional class, geometry, and the county characteristics that enter the formula, with standard errors clustered by county and wild-cluster-bootstrap inference given the small number of clusters.

Two constraints bind. The biennial survey rotation means that adding a second window requires a second even-year pavement vintage to pair with the 2024 survey of minor arterials and major collectors. That vintage is the 2026 survey, scheduled for release in January 2027; it completes a 2024-to-2026 window and would roughly double the cluster count from 29 to somewhere near 55 to 60. And county fixed effects remain unavailable, since funding is constant within county, so identification comes from comparisons across counties with the formula's own inputs as controls. Whether that is enough is the question the finished thesis must answer honestly; the present paper establishes only the prior that makes it interesting, which is that the allocation under which those funds arrive does not itself respond to need.

Data and code

The funding panel is constructed from public MDOT payment sheets by a Python parser I wrote for this project; it emits the panel, a reconciliation report for every county-month, and a quarantine file for line items it could not label. Pavement, traffic, and mileage inputs are public MDOT releases. Estimation is in Stata. All materials are available on request.

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